

Practice Week 2 Lecture Prepare Quiz: Decision Trees

- Due No due date
- Points 16
- Questions 16
- Time Limit None
- Allowed Attempts Unlimited

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Attempt History

	Attempt	Time	Score
KEPT	Attempt 2	3 minutes	16 out of 16
LATEST	Attempt 2	3 minutes	16 out of 16
	Attempt 1	710 minutes	11.67 out of 16

⚠ Correct answers are hidden.

Submitted Sep 21 at 10:58p.m.



Question 1

1 / 1 pts

What is a decision tree?

- ☒ A sequence of layered if-else statements used to make predictions on new data
- ☐ A method that always compares a test point to every training example in the dataset
- ☐ A model that can only be used for regression problems and not classification
- ☐ A random process for selecting class labels without using input features



Question 2

1 / 1 pts

What does a non-leaf node in a decision tree do?

- ☐ Aggregates the final predictions from its child nodes to produce an output
- ☒ Tests a condition on one feature to decide which child to follow
- ☐ Stores all training examples that reach the node for nearest-neighbor comparison
- ☐ Assigns a probability distribution over classes without making a split



Question 3

1 / 1 pts

What is a decision boundary?

- ☐ The set of training examples that are misclassified
- ☐ A set of lines that separates data perfectly based on their class
- ☒ A partition of the data space corresponding to predicted classes
- ☐ The maximum depth of a decision tree



Question 4

1 / 1 pts

What quantity is entropy used to measure?

- ☐ The number of nodes in a decision tree
- ☒ The uncertainty in a probability distribution
- ☐ The accuracy of k-nearest neighbours
- ☐ The amount of training data stored by a model



Question 5

1 / 1 pts

If a random variable has one outcome with probability 1, what is its entropy in bits?

- ☒ 0
- ☐ 1
- ☐ ∞
- ☐ $-\infty$



Question 6

1 / 1 pts

For a binary random variable Y with $P(Y = 1) = 0.75$ and $P(Y = 0) = 0.25$, which of the following correctly computes the entropy $H(Y)$ in bits?

- ☐ $H(Y) = -(0.75 \log_2 0.25 + 0.25 \log_2 0.75)$
- ☐ $H(Y) = 0.75 \log_2 0.75 + 0.25 \log_2 0.25$
- ☐ $H(Y) = 0.75 \log_2 0.25 + 0.25 \log_2 0.75$
- ☒ $H(Y) = -(0.75 \log_2 0.75 + 0.25 \log_2 0.25)$



Question 7

1 / 1 pts

What does conditional entropy measure?

- ☒ The average amount of uncertainty remaining in Y once the value of X is known
- ☐ The prior uncertainty of X before any information about Y has been collected
- ☐ The total reduction in the entropy of Y achieved by conditioning on X
- ☐ The combined entropy across variables X and Y simultaneously



Question 8

1 / 1 pts

Suppose $P(Y = 1 \mid X = x) = 0.9$ and $P(Y = 0 \mid X = x) = 0.1$. Which of the following correctly computes the conditional entropy $H(Y \mid X = x)$ in bits?

- ☐ $H(Y \mid X = x) = -(0.9 \log_2 0.1 + 0.1 \log_2 0.9)$
- ☐ $H(Y \mid X = x) = 0.9 \log_2 0.1 + 0.1 \log_2 0.9$
- ☒ $H(Y \mid X = x) = -(0.9 \log_2 0.9 + 0.1 \log_2 0.1)$
- ☐ $H(Y \mid X = x) = 0.9 \log_2 0.9 + 0.1 \log_2 0.1$



Question 9

1 / 1 pts

What is the information gain about random variable Y by observing random variable X defined as?

- ☐ $H(X) + H(Y)$
- ☐ $H(Y \mid X) - H(Y)$
- ☒ $H(Y) - H(Y \mid X)$
- ☐ $H(X, Y) - H(X)$



Question 10

1 / 1 pts

Information gain about random variable Y by observing random variable X is always nonnegative.

- ☒ True
- ☐ False



Question 11

1 / 1 pts

Decision tree decision boundaries are always axis-aligned because each split tests a threshold on a single feature.

- ☒ True
- ☐ False



Question 12

1 / 1 pts

What is a limitation of using training error alone as a split criterion in a decision tree?

- ☒ It ignores the remaining uncertainty in the resulting child nodes
- ☐ It can only be used to evaluate binary classification problems
- ☐ It requires all input features to support a differentiable error metric
- ☐ It cannot be applied when the input features take continuous values



Question 13

1 / 1 pts

What does the majority vote choose to predict?

- ☐ The class with the smallest entropy in the full dataset
- ☒ The most common class among the training points in that region
- ☐ A randomly chosen class from the region
- ☐ The class associated with the nearest training point



Question 14

1 / 1 pts

A greedy decision tree algorithm guarantees the globally optimal tree.

- ☐ True
- ☒ False



Question 15

1 / 1 pts

Why do decision tree algorithms only need to evaluate finitely many split thresholds for a continuous feature?

- ☒ Only thresholds between two observed values of the feature can change how the data points are partitioned
- ☐ Continuous features must first be converted into a small number of discrete categories before any splitting occurs
- ☐ Decision trees are limited to working with features that have a small and bounded range of values
- ☐ Every valid threshold must be chosen to exactly equal one of the feature values observed in the training data



Question 16

1 / 1 pts

Which of the following can be used as a stopping criterion during decision tree learning? Select all that apply.

- ☒ Stop splitting when the tree reaches a predefined maximum depth
- ☒ Stop splitting when a node contains fewer than a minimum number of samples
- ☒ Stop splitting when the best split produces too little information gain
- ☐ Stop splitting when all remaining features have different numeric scales